

Patan Multiple Campus

Mid-Term Examination-2082

Subject: Web Technology-II

Faculty: BIT

Semester: III/V

Time: 2:00 hrs.

F.M.: 40

P.M.: 16

Set B Group A

Long Answer Questions

ATTEMPT ANY ONE QUESTION. [$1 \times 10 = 10$]

1. What is a PHP Data Object? How does MySQLi Object Interface work? Write a PHP program to read data elements from a database table and display in a page. [2+3+5]
2. Consider you have a contact form with name, phone, and save/clear buttons. Write a PHP program to implement the contact form. The save button should redirect to success.php upon successful submission, and the clear button should reset the form fields. [10]

Group B

Short Answer Questions

ATTEMPT ANY SIX QUESTIONS. [$6 \times 5 = 30$]

3. Write a PHP program that will contain a join_strings function. How can you remove and insert elements in an array?. [5]
4. Write a PHP program to demonstrate the concept of inheritance. [5]
5. How can you validate form data in PHP? Illustrate with an example. [5]
6. Write a PHP program to split an array into chunks using array_split() function. [5]
7. Differentiate anonymous function from variable function. [5]
8. How can you combine PHP code with CSS in web pages? Write a PHP program to demonstrate the use of for and foreach loops. [5]
9. Write a PHP program to access properties and methods of a class. [5]

Patan Multiple Campus
Bachelor's in Information Technology (BIT)
Mid-Term Examination-2082

Section-A

Time: 2 hours

Subject: BIT 303 Information Security

Year/Sem: 3rd/V

F.M. : 40

P. M.: 16

Group "A"

Attempt Any One Questions:

[1×10=10]

1. What is Feistel Structure? Explain sub - key generation and round structure of DES algorithm with suitable diagram.
2. What is public key cryptography? Explain key generation in RSA algorithm. Discuss how encryption and decryption is done using RSA with suitable numerical example.

Group "B"

Attempt Any SIX questions:

[6×5=30]

3. What are different Security Design Principles? Explain any three of them in brief.
4. Encrypt and decrypt the message "NO" using the Hill Cipher with the key $\begin{pmatrix} 3 & 3 \\ 2 & 5 \end{pmatrix}$.
Show your calculations and result.
5. Describe how 128-bit of hash value is generated by taking an input message of variable size using MD5 algorithm?
6. What is vulnerability and threat? Explain different types of attack in brief.
7. Perform Primality test on 73 using Miller Rabin Test.
8. Explain Diffie Hellman Key Exchange Protocol with suitable example.
9. Write short notes on:
 - c. Attack Tree
 - d. Difference between Stream Cipher and Block Cipher

Patan Multiple Campus
Bachelor in Information Technology (BIT)
Mid-Term Exam-2082

Time: 2 hours
Semester: 5th

Subject: BIT304: Computer Graphics

Full Marks: 40

Pass Marks: 16

Section A(SET-A)

Group-A

Attempt any TWO questions. ($2 \times 10 = 20$)

1. List the major differences between DDA and Bradenham's Line drawing algorithm. Illustrate the DDA algorithm to the line width endpoints (2, 2) and (9, 2).
2. Let ABCD be the rectangle window with A(0, 0), B(10, 0), C(10, 10) and D(0, 10). Use Liang Barsky line clipping algorithm to clip the line XY where X(5, 3) and Y(15, 9)
3. Explain with Example how endpoint bits are calculated in Co-hen Sutherland line clipping algorithm.

Group-B

Attempt any FOUR questions. ($4 \times 5 = 20$)

4. Translate a triangle ABC with coordinates A(0, 0), B(5, 0), and C(5, 5) by 2 units in the x-direction and 3 units in the y-directions.
5. Differentiate between vector and raster scan systems. [5]
6. Find the new coordinate of the triangle ABC, with coordinates A(0,0), B(1,1), and C(5,2) after it has been magnified to twice of its size.
7. Explain how pivot point translation can be performed in 2D
8. Write short notes on [Any Two] [$2 \times 2.5 = 5$]
 - a. Bezier Curve and spline
 - b. Image Processing and Computer Graphics
 - c. 3D Object Representation

Tribhuvan University
Patan Multiple Campus
Bachelor in Information Technology (BIT)
Mid-Term Exam-2082

Set-B

Time: 2 hours (11:00 AM to 01:00 PM)
Subject: BIT302 Software Engineering
Semester: 5th

Total Marks: 40
Pass Marks: 16

Group - 'A'

Attempt any One Question.

[1 x 10 = 10]

1. Explain the use of graphical models in software engineering. [3+7]
2. What is requirements in software engineering? Explain the Functional and Non-functional requirement of software. [3+7]

Group - 'B'

Attempt any SIX Questions.

[6 x 5 = 30]

3. What views or perspectives are useful when designing and documenting a system's modeling? [5]
4. Explain the term Software engineering and System engineering? [5]
5. What is Software? Explain the attributes of good software? [5]
6. Explain Coping with change. What are two ways of coping with change and changing system requirements [5]
7. Explain the differences between Structural models and Behavioral models. [5]
8. Explain Agile Software Development method and its importance. Explain four key values of Agile Manifesto. [5]
9. What is Software validation? Explain the stages in the testing process. [5]

2, 3, 4, 5, 8, 7

Tribhuvan University
Patan Multiple Campus
Bachelor in Information Technology (BIT)

Mid-Term Examination- 2082

Subject: Technical Writing (ENG 305)

Semester: V

Time: 2 Hours

Total Marks: 40

Pass Marks: 20

Set-B(Section-A)

Group- A

Attempt any **TWO** questions. ($2 \times 10 = 20$)

1. Discuss the importance of technical writing for BIT students and professionals. Explain how technical writing skills contribute to career development.
2. Write a letter to the Minister of Education, the government of Nepal explaining why computer education should be made compulsory from class one.
3. Define intercultural communication and explain its importance in global workplaces.

Group- B

Attempt any **Four** questions. ($4 \times 5 = 20$)

1. Define technical sentences and explain their key characteristics.
2. What are the principles of designing effective technical graphics?
3. Differentiate between instructions and procedures.
4. How can a presenter handle nervousness and audience questions effectively?
5. Write an application for the post of IT Officer recently advertised in the Rising Nepal.